Autonomous Mobile Robot (AMR)

by Robotics Society

Abstract:

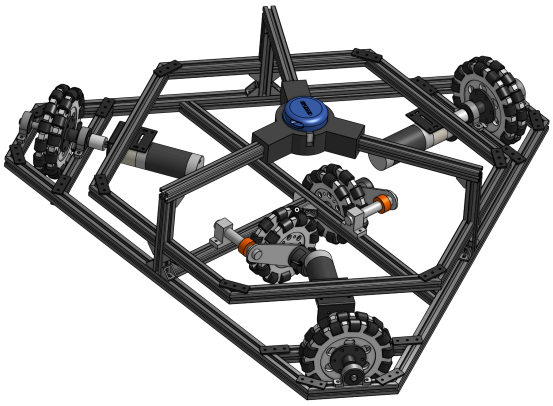
Our goal was to build a wheeled mobile robot that can autonomously navigate in an indoor environment. The key feature of it would be a rigid structure that can carry a lot of weight, have flexible movement and behave dynamically to its surroundings.

Approach:

We went for a design that implements 3 omni-wheels allowing holonomic motion at minimum cost.
A rigid and precise structure was possible by use of 20x20 Aluminum Extrusion channels.
A Local Computer running ROS2, navigation stack of Nav2 and using sensor data of a Lidar (Visual odometry), rotary encoders (Absolute Odometry) and IMU (Inertial Odometry) we were able to make it navigate autonomously through waypoints.

Design:

The CAD design with the final materials was prepared to estimate the requirements and prepare a BOM.
Design Patent Certificate



Applications:

This design acts as a base frame for proof of concept. With further extensions and tweaks in design it can have a wide range of applications

. Hospitality Sector - Waiter Robot for Serving at restaurants, hotels, lounges
. As Cleaning robot at public spaces such as malls, railway platforms, airports.
. Carrying small loads such as luggages in railway platforms or airports, bus bays or from door to door service. Applicable in small Industries too.

Future Aspects/Goals:

Achieving higher accuracy and precision in the current design would be the first priority. Simultaneously some other formats can be developed such as ones with 4 mecanum wheels or differential drive mechanism.
Development of a User interface for general purpose usage would be beneficial.

Project By: Shanti Sworup Biswal, Subham Mishra, Soumyajit Pattnaik, Parthasarathi Swain

Pictures & Videos

